



4.7 Bayes' theorem (HL)

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Bayes' theorem allows reversal of conditional probabilities: $P(A|B) = P(B|A) \cdot P(A) / P(B)$, where $P(B)$ is often found using the law of total probability, $P(B) = P(B|A)P(A) + P(B|A')P(A')$. This is used to update the probability of a cause/event A given new evidence B , and is commonly represented and computed using a tree diagram, reading branches forward for $P(B|A)$ and combining to get $P(A|B)$.

Worked example: follow the reasoning

A test for a disease is 95% accurate for those with the disease (true positive) and 90% accurate for those without (true negative). The disease has prevalence 2% in the population. If a person tests positive, find the probability they have the disease.

$P(D)=0.02$, $P(D')=0.98$. $P(+|D)=0.95$, $P(+|D')=0.10$.

$P(+) = 0.95(0.02) + 0.10(0.98) = 0.019 + 0.098 = 0.117$.

$P(D|+) = P(+|D)P(D)/P(+) = 0.019/0.117 = 0.1624 \approx 0.162$.

Quick check · multiple choice

Bayes' theorem is primarily used to:

A Find the union of two events

B Reverse a conditional probability using known conditional probabilities

C Test for independence of two events

D Calculate the mean of a distribution

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Box A contains 3 red and 7 blue balls; Box B contains 6 red and 4 blue balls. A box is chosen at random (equal chance) and one ball is drawn. [6]
- (a) Find the probability that the ball drawn is red.
- (b) Given that the ball drawn is red, find the probability it came from Box A.

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $P(\text{Red}) = P(A)P(\text{Red}|A) + P(B)P(\text{Red}|B) = 0.5(0.3) + 0.5(0.6) = 0.15 + 0.3 = 0.45.$

(b) $P(A|\text{Red}) = P(\text{Red}|A)P(A)/P(\text{Red}) = (0.3 \times 0.5)/0.45 = 0.15/0.45 = 1/3.$

Quick check answer

B. Bayes' theorem computes $P(A|B)$ from $P(B|A)$ and the prior probabilities.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.